

Digital Signal Processing

Detailed Exam Solutions (Part I - 2024)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for all questions in **PART I** of the Digital Signal Processing exam paper (Ref No: Ex/EE/PC/B/T/224/2024).

Question 1

(a) Nyquist Rate and Spectral Components

Problem: Determine the Nyquist rate for $x(t) = 2 \cos^2(2\pi t) \sin(6\pi t) + 3 \sin(16\pi t)$. If $x(t)$ is sampled at 4 Hz, and passed through an ideal low-pass filter with cutoff 2 Hz, determine the frequencies of the spectral components at the output.

Solution:

1. Find the Nyquist Rate:

Expand the first term using $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$:

$$\begin{aligned} x_1(t) &= 2 \left(\frac{1 + \cos(4\pi t)}{2} \right) \sin(6\pi t) = (1 + \cos(4\pi t)) \sin(6\pi t) \\ &= \sin(6\pi t) + \cos(4\pi t) \sin(6\pi t) \end{aligned}$$

Using the product-to-sum formula $\cos(A) \sin(B) = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$:

$$\begin{aligned} x_1(t) &= \sin(6\pi t) + \frac{1}{2}[\sin(10\pi t) - \sin(-2\pi t)] \\ &= \sin(6\pi t) + \frac{1}{2} \sin(10\pi t) + \frac{1}{2} \sin(2\pi t) \end{aligned}$$

Combining with the second term, the full signal is:

$$x(t) = \frac{1}{2} \sin(2\pi t) + \sin(6\pi t) + \frac{1}{2} \sin(10\pi t) + 3 \sin(16\pi t)$$

The continuous-time frequencies $f_i = \frac{\omega_i}{2\pi}$ are: $f_1 = 1$ Hz, $f_2 = 3$ Hz, $f_3 = 5$ Hz, $f_4 = 8$ Hz.

The maximum frequency component is $f_{max} = 8$ Hz.

Nyquist rate: $f_{Nyquist} = 2f_{max} = 16$ Hz.

2. Reconstructed Frequencies:

The signal is sampled at $F_s = 4$ Hz. Since $4 \text{ Hz} < 16 \text{ Hz}$, aliasing occurs. The aliased frequency f_a is $f_a = |f_{in} - kF_s|$ where k is an integer such that $0 \leq f_a \leq F_s/2 = 2$ Hz.

- $f_1 = 1$ Hz: $k = 0 \implies f_a = |1 - 0| = 1$ Hz.
- $f_2 = 3$ Hz: $k = 1 \implies f_a = |3 - 4| = 1$ Hz.

- $f_3 = 5 \text{ Hz}$: $k = 1 \implies f_a = |5 - 4| = 1 \text{ Hz}$.
- $f_4 = 8 \text{ Hz}$: $k = 2 \implies f_a = |8 - 8| = 0 \text{ Hz (DC)}$.

The sampled signal contains components at 0 Hz (DC) and 1 Hz. An ideal low-pass filter with a cutoff of 2 Hz will pass both of these components.

Final Output Frequencies: 0 Hz and 1 Hz.

(b) Anti-aliasing Filter

Problem: Explain the necessity of an analog lowpass filter prior to an ADC. How is its cutoff selected? Give a practical example.

Solution:

Necessity: The analog lowpass filter placed before an ADC is called an **anti-aliasing filter**. Its necessity stems from the Nyquist-Shannon sampling theorem, which dictates that to perfectly reconstruct a signal without aliasing distortion, the sampling rate F_s must be strictly greater than twice the highest frequency present in the signal. Real-world analog signals often contain high-frequency noise or unwanted harmonics out to infinity. If sampled directly, these high frequencies will fold back (alias) into the baseband, irreversibly corrupting the digital signal. The anti-aliasing filter forcefully limits the bandwidth of the continuous signal to prevent this.

Selection of Cutoff Frequency (f_c): The cutoff frequency f_c should be chosen to be slightly higher than the maximum frequency of interest (f_{max_signal}), but strictly less than or equal to the folding frequency ($F_s/2$). Practically, because ideal brick-wall filters don't exist, a guard band is left. Thus, $f_{max_signal} < f_c < F_s/2$.

Practical Example: In a standard audio CD system, human hearing ranges up to approximately 20 kHz (so $f_{max_signal} = 20 \text{ kHz}$). The sampling frequency is chosen as $F_s = 44.1 \text{ kHz}$. The folding frequency is $44.1/2 = 22.05 \text{ kHz}$. The cutoff frequency of the anti-aliasing filter is typically chosen around $f_c \approx 20.5 \text{ kHz}$ to pass the audible range while steeply rolling off before 22.05 kHz to prevent any aliasing of ultrasonic noise.

Question 2

(a) Complex-conjugate Poles in Z-plane

Problem: Cite a sequence whose Z-transform has a complex-conjugate pair of poles. Examine the effect of pole location on the unit circle, outside, and inside the unit circle, and the time-domain behavior.

Solution:

A standard causal sequence that yields a complex-conjugate pair of poles is a damped sinusoid:

$$x[n] = r^n \cos(\omega_0 n) u[n] \quad \text{for } r > 0$$

Its Z-transform is:

$$X(z) = \frac{1 - r \cos(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$$

The poles are located at: $p_{1,2} = r e^{\pm j\omega_0}$. This is a complex-conjugate pair with magnitude $|p_{1,2}| = r$ and angle $\pm\omega_0$.

Effect of Pole Location:

1. **Inside the Unit Circle ($r < 1$):** The poles are located strictly inside the unit circle. The time-domain sequence $x[n] = r^n \cos(\omega_0 n)$ exponentially decays as $n \rightarrow \infty$. The system is stable because the impulse response is absolutely summable.
2. **On the Unit Circle ($r = 1$):** The poles are located exactly on the circumference of the unit circle. The sequence becomes $x[n] = \cos(\omega_0 n)$, which is a steady, un-damped oscillation. The system is marginally stable.
3. **Outside the Unit Circle ($r > 1$):** The poles are located outside the unit circle. The time-domain sequence grows exponentially without bound as $n \rightarrow \infty$. The system is strictly unstable.

(b) Z-transform Properties

Problem: Obtain the closed form expression for the Z-transform of a unit step sequence. Use properties to find the transform of $h[n] = n(\frac{1}{2})^n u[n-2]$. State ROCs and poles.

Solution:

1. Unit Step Sequence:

$u[n] = 1$ for $n \geq 0$.

$$Z\{u[n]\} = \sum_{n=0}^{\infty} (1) z^{-n} = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

ROC: $|z| > 1$. **Pole:** $z = 1$ (multiplicity 1).

2. Transform of $h[n] = n(\frac{1}{2})^n u[n-2]$:

Let $x_1[n] = (\frac{1}{2})^n u[n]$. Its transform is $X_1(z) = \frac{1}{1 - 0.5z^{-1}}$ with ROC $|z| > 0.5$.

By the multiplication by n property (differentiation in z-domain):

$$Z\{n \cdot x_1[n]\} = -z \frac{d}{dz} X_1(z) = -z \frac{d}{dz} \left(\frac{z}{z - 0.5} \right) = -z \left(\frac{1(z - 0.5) - z(1)}{(z - 0.5)^2} \right) = \frac{0.5z}{(z - 0.5)^2}$$

Let $x_2[n] = n(\frac{1}{2})^n u[n]$. Note that for $n = 0$ and $n = 1$, $x_2[0] = 0$ and $x_2[1] = 0.5$. The sequence we want is $h[n] = n(\frac{1}{2})^n u[n-2]$. Notice that $h[n] = x_2[n]$ for all $n \geq 2$. But at $n = 1$, $x_2[1] = 0.5$

while $h[1] = 0$. Therefore, $h[n] = x_2[n] - 0.5\delta[n-1]$. Taking the Z-transform:

$$\begin{aligned} H(z) &= Z\{x_2[n]\} - 0.5Z\{\delta[n-1]\} \\ &= \frac{0.5z}{(z-0.5)^2} - 0.5z^{-1} = \frac{0.5z^{-1}}{(1-0.5z^{-1})^2} - 0.5z^{-1} \end{aligned}$$

Alternatively, let's write $h[n]$ explicitly by shifting: Let $m = n - 2 \implies n = m + 2$.

$$\begin{aligned} h[n] &= (m+2)\left(\frac{1}{2}\right)^{m+2}u[m] = \frac{1}{4}(m+2)\left(\frac{1}{2}\right)^m u[m] \\ &= \frac{1}{4} \left[m\left(\frac{1}{2}\right)^m u[m] + 2\left(\frac{1}{2}\right)^m u[m] \right] \end{aligned}$$

Transform:

$$H(z) = \frac{1}{4} \left[\frac{0.5z^{-1}}{(1-0.5z^{-1})^2} + 2\frac{1}{1-0.5z^{-1}} \right] z^{-2} \quad \text{WAIT, shifting handles the } z^{-2}.$$

Actually, if $h[n] = (n)\left(\frac{1}{2}\right)^n u[n-2]$, simply using the direct definition: $H(z) = \sum_{n=2}^{\infty} n\left(\frac{1}{2}\right)^n z^{-n}$. Recall $\sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-az^{-1}}$. Differentiating w.r.t a : $\sum_{n=0}^{\infty} na^{n-1} z^{-n} = \frac{z^{-1}}{(1-az^{-1})^2} \implies \sum_{n=0}^{\infty} na^n z^{-n} = \frac{az^{-1}}{(1-az^{-1})^2}$. The sum from $n=2$ misses $n=0$ (which is 0) and $n=1$ (which is az^{-1}). $H(z) = \frac{az^{-1}}{(1-az^{-1})^2} - az^{-1}$. For $a = 0.5$:

$$H(z) = \frac{0.5z^{-1}}{(1-0.5z^{-1})^2} - 0.5z^{-1}$$

To express as a single fraction:

$$H(z) = \frac{0.5z^{-1} - 0.5z^{-1}(1 - z^{-1} + 0.25z^{-2})}{(1-0.5z^{-1})^2} = \frac{0.5z^{-2} - 0.125z^{-3}}{(1-0.5z^{-1})^2} = \frac{z(0.5) - 0.125}{z(z-0.5)^2}$$

ROC: $|z| > 0.5$.

Poles: $z = 0.5$ (multiplicity 2) and $z = 0$ (multiplicity 1).

(c) Inverse Z-Transform

Problem: Determine the closed-form expression for the inverse z-transform of $F(z) = \frac{z}{z-0.5}$; $|z| > 0.5$.

Solution:

Rewrite $F(z)$ in powers of z^{-1} :

$$F(z) = \frac{1}{1-0.5z^{-1}}$$

The ROC is given as $|z| > 0.5$, which represents the exterior of a circle of radius 0.5 in the z-plane. An outward-pointing ROC corresponds to a right-sided (causal) sequence. Using the standard Z-transform pair $a^n u[n] \xrightarrow{Z} \frac{1}{1-az^{-1}}$ for $|z| > |a|$:

$$f[n] = (0.5)^n u[n]$$

Question 3

(a) Pole-Zero Configuration and System Properties

Problem: A DTLTI system has poles at $z = 0.7, 1.1$ and a zero at $z = -0.4$. (Assuming based on typical problem structures matching the OCR values 0.7, 1.1, -0.4). DC gain is 9.5. Obtain $H(z)$, identify ROCs, causality, stability, and impulse response.

Solution:

1. Transfer Function:

The transfer function with poles $p_1 = 0.7$, $p_2 = 1.1$ and zero $z_1 = -0.4$ is:

$$H(z) = K \frac{z - z_1}{(z - p_1)(z - p_2)} = K \frac{z + 0.4}{(z - 0.7)(z - 1.1)}$$

The DC gain is defined at $z = 1$. $H(1) = 9.5$:

$$9.5 = K \frac{1 + 0.4}{(1 - 0.7)(1 - 1.1)} = K \frac{1.4}{(0.3)(-0.1)} = K \frac{1.4}{-0.03}$$

$$K = \frac{9.5 \times (-0.03)}{1.4} = -0.20357$$

$$H(z) = \frac{-0.20357(z + 0.4)}{(z - 0.7)(z - 1.1)}$$

2. ROCs, Stability, and Causality:

The poles are at radii 0.7 and 1.1. This divides the z -plane into 3 possible ROCs:

- **ROC 1:** $|z| > 1.1$. The sequence is right-sided (**causal**). The ROC does not include the unit circle $|z| = 1$, so the system is **unstable**.
- **ROC 2:** $0.7 < |z| < 1.1$. The sequence is two-sided (**non-causal**). The ROC includes the unit circle, so the system is **stable**.
- **ROC 3:** $|z| < 0.7$. The sequence is left-sided (**anti-causal**). The ROC does not include the unit circle, so the system is **unstable**.

3. Impulse Response for Causal System:

If the system is causal, ROC is $|z| > 1.1$. We use Partial Fraction Expansion on $H(z)/z$:

$$\frac{H(z)}{z} = \frac{-0.20357(z + 0.4)}{z(z - 0.7)(z - 1.1)} = \frac{A}{z} + \frac{B}{z - 0.7} + \frac{C}{z - 1.1}$$

Solving for coefficients:

- $A = \frac{-0.20357(0.4)}{(-0.7)(-1.1)} = -0.1057$
- $B = \frac{-0.20357(0.7+0.4)}{0.7(0.7-1.1)} = \frac{-0.20357(1.1)}{0.7(-0.4)} = \frac{-0.2239}{-0.28} = 0.7997$
- $C = \frac{-0.20357(1.1+0.4)}{1.1(1.1-0.7)} = \frac{-0.20357(1.5)}{1.1(0.4)} = \frac{-0.3053}{0.44} = -0.6940$

$$H(z) = -0.1057 + \frac{0.7997z}{z - 0.7} - \frac{0.6940z}{z - 1.1}$$

Taking the inverse Z-transform for a causal ROC:

$$h[n] = -0.1057\delta[n] + 0.7997(0.7)^n u[n] - 0.6940(1.1)^n u[n]$$

(b) IIR System Difference Equation

Problem: $y[n] = bx[n] + ay[n-1]$. Show it's IIR. Examine effect of a and b on impulse response.

Solution:

To find the impulse response $h[n]$, let $x[n] = \delta[n]$, assuming initial rest $y[-1] = 0$.

$$h[0] = b\delta[0] + ah[-1] = b(1) + a(0) = b$$

$$h[1] = b\delta[1] + ah[0] = 0 + a(b) = ab$$

$$h[2] = b\delta[2] + ah[1] = 0 + a(ab) = a^2b$$

...

$$h[n] = b \cdot a^n \quad \text{for } n \geq 0$$

Thus, $h[n] = b \cdot a^n u[n]$. **IIR Nature:** Since $h[n]$ is defined for all $n \geq 0$ out to infinity (as long as $a \neq 0$), the impulse response has an infinite number of non-zero terms. Therefore, it is an Infinite Impulse Response (IIR) system.

Effect of a and b :

- ****Effect of b :** The parameter b strictly acts as an initial scaling factor (or gain). It determines the starting amplitude of the impulse response ($h[0] = b$).
- ****Effect of a :** The parameter a defines the pole of the system and dictates the dynamic behavior.
 - If $0 < a < 1$: $h[n]$ is a positive, monotonically decaying exponential (stable).
 - If $-1 < a < 0$: $h[n]$ is an alternating-sign decaying exponential (stable).
 - If $a > 1$: $h[n]$ is a positive exponentially growing sequence (unstable).
 - If $a < -1$: $h[n]$ is an alternating-sign exponentially growing sequence (unstable).
 - If $a = 1$: $h[n]$ is a constant step sequence (marginally stable).

Question 4

Butterworth Filter Design using Bilinear Transform

Problem: Low pass Butterworth filter, DC gain 0 dB, $A_p = 3$ dB at 675 Hz, $A_s \geq 15$ dB at 1350 Hz. Bilinear with pre-warping. $F_s = 10$ kHz. Obtain difference equation, prove warping formula, and explain characteristic aliasing absence.

Solution:

1. Analog Frequencies & Pre-warping:

Digital frequencies: $\omega_p = \frac{2\pi(675)}{10000} = 0.4241$ rad/sample. $\omega_s = \frac{2\pi(1350)}{10000} = 0.8482$ rad/sample.
Pre-warped analog frequencies ($\Omega = 2F_s \tan(\omega/2)$):

$$\Omega_p = 20000 \tan(0.21205) = 4305.89 \text{ rad/s}$$

$$\Omega_s = 20000 \tan(0.4241) = 9030.35 \text{ rad/s}$$

2. Filter Order (N):

$$N \geq \frac{\log_{10} \sqrt{(10^{0.1(15)} - 1)/(10^{0.1(3)} - 1)}}{\log_{10}(\Omega_s/\Omega_p)} = \frac{\log_{10} \sqrt{30.62/0.995}}{\log_{10}(2.097)} = 2.313 \implies N = 3$$

Let $\Omega_c = \Omega_p = 4305.89$ rad/s (since $A_p \approx 3$ dB).

3. Analog Filter $H(s)$: For a 3rd-order Butterworth, poles are at angles $\pm 120^\circ, 180^\circ$ on a circle of radius Ω_c . The denominator polynomial is $s^3 + 2\Omega_c s^2 + 2\Omega_c^2 s + \Omega_c^3$.

$$H(s) = \frac{\Omega_c^3}{s^3 + 2\Omega_c s^2 + 2\Omega_c^2 s + \Omega_c^3}$$

4. Bilinear Transformation: Substitute $s = 2F_s \frac{1-z^{-1}}{1+z^{-1}} = 20000 \frac{1-z^{-1}}{1+z^{-1}}$ into $H(s)$. After algebraic expansion and substituting $\Omega_c = 4305.89$, we get $H(z)$:

$$H(z) = \frac{0.0065 + 0.0195z^{-1} + 0.0195z^{-2} + 0.0065z^{-3}}{1 - 2.1574z^{-1} + 1.6348z^{-2} - 0.4253z^{-3}}$$

(Normalized so $a_0 = 1$ and DC gain $H(1) = 1$).

Difference Equation:

$$y[n] = 2.1574y[n-1] - 1.6348y[n-2] + 0.4253y[n-3] + 0.0065x[n] + 0.0195x[n-1] + 0.0195x[n-2] + 0.0065x[n-3]$$

5. Proof of Warping Formula: The Bilinear transform maps the $j\Omega$ axis to the unit circle $e^{j\omega}$:

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \implies j\Omega = \frac{2}{T} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}}$$

Factoring out half-angles:

$$j\Omega = \frac{2}{T} \frac{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2}(e^{j\omega/2} + e^{-j\omega/2})} = \frac{2}{T} \frac{2j \sin(\omega/2)}{2 \cos(\omega/2)} = j \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

Thus, $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$, proving the non-linear relationship (warping).

6. Why it is free from Characteristic Aliasing: The Bilinear Transform compresses the **entire** infinite continuous imaginary axis $-\infty < \Omega < \infty$ exactly onto a single revolution of the discrete unit circle $-\pi < \omega < \pi$. Since no two points on the s -plane's imaginary axis map to the same point on the z -plane's unit circle, there is a one-to-one mapping. Consequently, the high-frequency tails of the analog filter do not overlap or fold back into the baseband, completely avoiding characteristic aliasing.

Question 5 (Short Notes)

(a) Designing digital filters by impulse invariant technique:

The impulse invariant technique designs IIR digital filters by sampling the impulse response of an analog prototype filter: $h[n] = T_d h_a(nT_d)$. This maintains the time-domain impulse characteristics. However, sampling in the time domain causes periodic replication in the frequency domain. If the analog filter is not strictly bandlimited, these replications overlap, causing **aliasing**. It is best suited for low-pass and band-pass filters with sharp cutoffs, and cannot be used for high-pass filters.

(b) Digital Integrators:

A digital integrator approximates continuous integration in discrete time. The continuous relation $y(t) = \int x(t)dt$ has $H(s) = 1/s$. Applying transformation techniques yields various digital integrators. The Bilinear transform (Trapezoidal rule) yields $H(z) = \frac{T}{2} \frac{1+z^{-1}}{1-z^{-1}}$. The Forward Euler yields $H(z) = \frac{Tz^{-1}}{1-z^{-1}}$, and Backward Euler yields $H(z) = \frac{T}{1-z^{-1}}$. They are fundamental building blocks in digital control and signal processing systems.

(c) Mapping of left half of s-plane on to z-plane:

In the standard $z = e^{sT}$ mapping, substituting $s = \sigma + j\Omega$ gives $z = e^{\sigma T} e^{j\Omega T}$. The magnitude is $|z| = e^{\sigma T}$. The left half of the s-plane corresponds to $\sigma < 0$. If $\sigma < 0$, then $e^{\sigma T} < 1$, meaning $|z| < 1$. Thus, the entire stable left half of the analog s-plane maps to the interior of the unit circle in the digital z-plane, preserving stability.